

Type juggling in PHP

It means PHP automatically changes a value from one data type to another when it thinks it is needed.

PHP has several basic types:

- `int` → whole numbers (10)
- `float` → decimals (3.14)
- `string` → text ("hello")
- `bool` → true/false (true)
- array, object, etc.

Example 1: String becomes a number

```
<?php
$x = "10";
$y = 5;

echo $x + $y;
?>
```

Output:

15

What happened?

"10" + 5

PHP sees + (math operation), so it converts:

"10" → 10

Then:

10 + 5 = 15

Example 2: Number becomes a string

```
<?php
$age = 25;

echo "I am " . $age . " years old";
?>
```

Output:

I am 25 years old

The `.` operator joins strings, so PHP converts:

`25` → `"25"`

Result:

`"I am " + "25" + " years old"`

Example 3: Boolean conversion

PHP also converts values to `true` or `false` when needed.

```
<?php
$name = "";

if ($name) {
    echo "Hello";
} else {
    echo "No name";
}
?>
```

Output:

No name

Why?

An empty string is considered **false**:

`"` → `false`

Some common false-like values:

```
false
0
0.0
""
"0"
null
[]
```

Most other values are considered true.

Type juggling with comparisons

This is where PHP can surprise people.

Loose comparison (= =)

```
var_dump(5 == "5");
```

Output:

```
true
```

PHP converts "5" into 5:

```
5 == 5
```

Strict comparison (===)

```
var_dump(5 === "5");
```

Output:

```
false
```

Because strict comparison checks both:

1. Value
2. Type

So:

```
5           → integer  
"5"        → string
```

They are different.

A common beginner surprise

```
var_dump("10cats" + 5);
```

PHP may convert the beginning numeric part:

```
"10cats" → 10
```

Then:

```
10 + 5 = 15
```

This behavior can lead to bugs, so modern PHP code often uses explicit conversions.

Avoiding surprises: explicit casting

Instead of letting PHP guess:

```
$number = "100";  
  
$total = (int)$number + 20;  
  
echo $total;
```

Here you are saying:

"100" → convert to integer

Result:

120

Simple rule of thumb

Situation	PHP may do
Math with strings	Convert strings to numbers
String joining	Convert numbers to strings
if conditions	Convert values to Boolean
== comparison	Convert types to match
=== comparison	Do not convert types

In short: Type juggling is PHP automatically changing types to make an operation work. It is convenient, but using strict comparisons (===) and explicit casts helps prevent unexpected behavior.